MAPS FACTORING THROUGH A PROJECTIVE MODULE FORM A SUBMODULE AND ARE STABLE UNDER COMPOSITION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Projective module, characterised as a direct summand of a free module. the author cites Algebra, Lemma algebra-lemma-characterize-projective at proof line 17726. Supplies the equivalence in the first part.
- Direct sums of projective modules are projective, and a map into a direct sum factors through its components. used at proof lines 17728-17732. Gives additivity for the second part.
- Composition of maps and the associativity of the factorisation through a fixed module. used at proof lines 17732-17735. Gives the stability statement of the third part.

Source reference. Original lemma 0G90: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20285–20285. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 05CF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6918–6918. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let M, N be R-modules.*

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) *Given an R-module map $\varphi : M \to N$ the following are equivalent: (a) φ factors through a projective R-module, and (b) φ factors through a free R-module.*
(2) *The set of $\varphi : M \to N$ satisfying the equivalent conditions of (1) is an R-submodule of $\mathrm{Hom}_R(M, N)$.*
(3) *Given maps $\psi : M' \to M$ and $\xi : N \to N'$, if $\varphi : M \to N$ satisfies the equivalent conditions of (1), then $\xi \circ \varphi \circ \psi : M' \to N'$ does too.*

Proof. The equivalence of (1)(a) and (1)(b) follows from Algebra, Lemma 05CF. If $\varphi : M \to N$ and $\varphi' : M \to N$ factor through the modules P and P' then $\varphi + \varphi'$ factors through $P \oplus P'$ and $\lambda\varphi$ factors through P for all $\lambda \in R$. This proves (2). If $\varphi : M \to N$ factors through the module P and ψ and ξ are as in (3), then $\xi \circ \varphi \circ \psi$ factors through P. This proves (3). □